



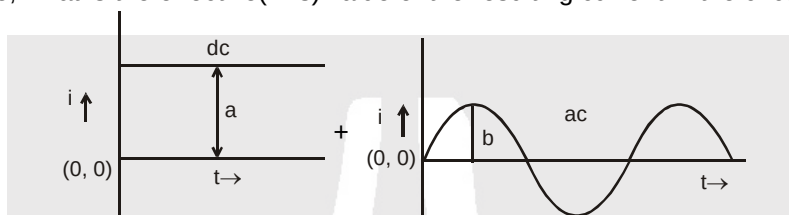
Exercise-1

Marked Questions can be used as Revision Questions.

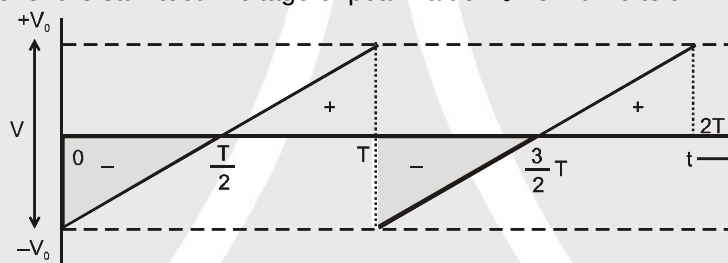
PART - I : SUBJECTIVE QUESTIONS

Section (A) : Average, peak and RMS value

- A-1.** The household supply of electricity is at 220V rms value and 50 Hz. Calculate the peak voltage and the minimum possible time in which the voltage can change from the rms value to zero.
- A-2.** In a LR circuit discharging current is given by $I = I_0 e^{-t/\tau}$ where τ is the time constant of the circuit find the rms current for the period $t = 0$ to $t = \tau$.
- A-3.** If a direct current of value 'a' ampere is superimposed on an alternating current $I = b \sin \omega t$ flowing through a wire, what is the effective(rms) value of the resulting current in the circuit?



- A-4.** Find the average for the saw-tooth voltage of peak value V_0 from $t = 0$ to $t = 2T$ as shown in figure.



Section (B) : Power consumed in an ac circuit

- B-1.** A bulb is designed to operate at 12 volts constant direct current. If this bulb is connected to an alternating current source and gives same brightness. What would be the peak voltage of the source ?
- B-2.** A resistor of resistance 100Ω is connected to an AC source $\varepsilon = (12V) \sin (250\pi s^{-1})t$. Find the power consumed by the bulb.
- B-3.** In an ac circuit the instantaneous values of current and applied voltage are respectively $i = 2(\text{Amp}) \sin (250 \pi s^{-1})t$ and $\varepsilon = (10V) \sin [(250 \pi s^{-1})t + \frac{\pi}{3}]$. Find the instantaneous power drawn from the source at $t = \frac{2}{3} \text{ ms}$ and its average value.

Section (C) : AC source with R, L, C connected in series

- C-1.** The dielectric strength of air is $3.0 \times 10^6 \text{ V/m}$. A parallel plate air capacitor has area 20 cm^2 and plate separation $\sqrt{2} \text{ mm}$. Find the maximum rms voltage of an AC source which can be safely connected to this capacitor.
- C-2.** An electric bulb is designed to consume 55 W when operated at 110 volts. It is connected to a 220 V, 50 Hz line through a choke coil in series. What should be the inductance of the coil for which the bulb gets correct voltage ?



- C-3.** A resistor, a capacitor and an inductor ($R = 300 \Omega$, $C = 20 \mu\text{F}$, $L = 1.0 \text{ henry}$) are connected in series with an AC source of, $E_{\text{rms}} = 50 \text{ V}$ and $\omega = \frac{50}{\pi} \text{ Hz}$. Find (a) the rms current in the circuit and (b) the rms potential differences across the capacitor, the resistor and the inductor.
- C-4.** Consider the situation of the previous problem calculate the average electric field energy stored in the capacitor and the average magnetic field energy stored in the inductor coil.
- C-5.** A 20 volts 5 watt lamp (lamp to be treated as a resistor) is used on AC mains of 200 volts and $\frac{50}{\pi} \sqrt{11}$ c.p.s. Calculate the (i) capacitance of the capacitor, or inductance of the inductor, to be put in series to run the lamp. (ii) How much pure resistance should be included in place of the above device so that the lamp can run on its rated voltage. (iii) which is more economical (the capacitor, the inductor or the resistor).
- C-6.** A circuit has a resistance of 50 ohms and an inductance of $\frac{3}{\pi}$ henry. It is connected in series with a condenser of $\frac{40}{\pi} \mu\text{F}$ and AC supply voltage of 200 V and 50 cycles/sec. Calculate
(i) the impedance of the circuit,
(ii) the p.d. across inductor coil and condenser.
(iii) Power factor
- C-7.** A coil draws a current of 1.0 ampere and a power of 100 watt from an A.C. source of 110 volt and $\frac{5\sqrt{22}}{\pi}$ hertz. Find the inductance and resistance of the coil.
- C-8.** An inductor $2/\pi$ Henry, a capacitor $100/\pi \mu\text{F}$ and a resistor 75Ω are connected in series across a source of emf $V = 10 \sin 100 \pi t$. Here t is in second. (a) find the impedance of the circuit. (b) find the energy dissipated in the circuit in 20 minutes.

Section (D) : Resonance

- D-1.** A series circuit consists of a resistance, inductance and capacitance. The applied voltage and the current at any instant are given by $E = 141.4 \cos (5000 t - 10^\circ)$ and $I = 5 \cos (5000 t - 370^\circ)$. The inductance is 0.01 henry. Calculate the value of capacitance and resistance.
- D-2.** An inductance of 2.0 H, a capacitance of $18 \mu\text{F}$ and a resistance of $10 \text{ k}\Omega$ are connected to an AC source of 20V with adjustable frequency (a) What frequency should be chosen to maximise the current(RMS) in the circuit? (b) What is the value of this maximum current (RMS) ?
- D-3.** An inductor-coil, a capacitor are connected in series with an AC source of rms voltage 24 V. When the frequency of the source is varied a maximum rms current of 6.0 A is observed. If this inductor coil is connected to a DC source of 12 V and having internal resistance 4.0Ω , what will be the current in steady state?
- D-4.** An electro magnetic wave of wavelength 300 metre can be transmitted by a transmission centre. A condenser of capacity $2.5 \mu\text{F}$ is available. Calculate the inductance of the required coil for a resonant circuit. Use $\pi^2 = 10$.

Section (E) : Transformer

- E-1** A transformer has 50 turns in the primary and 100 turns in the secondary. If the primary is connected to a 220 V DC supply, what will be the voltage across the secondary ?
- E-2.** In a transformer ratio of secondary turns (N_2) and primary turns (N_1) i.e. $\frac{N_2}{N_1} = 4$. If the voltage applied in primary is 200 V, 50 Hz, find (a) voltage induced in secondary (b) If current in primary is 1A, find the current in secondary if the transformer is (i) ideal and (ii) 80% efficient and there is no flux leakage.



PART - II : ONLY ONE OPTION CORRECT TYPE

Section (A) : Average, peak and RMS values and RMS values

- A-1.** r.m.s. value of current $i = 3 + 4 \sin(\omega t + \pi/3)$ is:
- (A) 5 A (B) $\sqrt{17}$ A (C) $\frac{5}{\sqrt{2}}$ A (D) $\frac{7}{\sqrt{2}}$ A
- A-2.** The peak value of an alternating e.m.f given by $E = E_0 \cos \omega t$, is 10 volt and frequency is 50 Hz. At time $t = (1/600)$ sec, the instantaneous value of e.m.f is :
- (A) 10 volt (B) $5\sqrt{3}$ volt (C) 5 volt (D) 1 volt
- A-3.** The voltage of an AC source varies with time according to the equation, $V = 100 \sin 100 \pi t \cos 100 \pi t$. Where t is in second and V is in volt. Then :
- (A) the peak voltage of the source is 100 volt (B) the peak voltage of the source is $(100/\sqrt{2})$ volt
(C) the peak voltage of the source is 50 volt (D) the frequency of the source is 50 Hz
- A-4.** An alternating voltage is given by : $e = e_1 \sin \omega t + e_2 \cos \omega t$. Then the root mean square value of voltage is given by :
- (A) $\sqrt{e_1^2 + e_2^2}$ (B) $\sqrt{e_1 e_2}$ (C) $\sqrt{\frac{e_1^2 + e_2^2}{2}}$ (D) $\sqrt{\frac{e_1^2 + e_2^2}{2}}$
- A-5.** An AC voltage is given by $E = E_0 \sin \frac{2\pi t}{T}$. Then the mean value of voltage calculated over time interval of $T/2$ seconds :
- (A) is always zero (B) is never zero (C) is $(2E_0/\pi)$ always (D) may be zero
- A-6.** An AC voltage of $V = 220\sqrt{2} \sin \left(100\pi t + \frac{\pi}{2} \right)$ is applied across a DC voltmeter, its reading will be:
- (A) $220\sqrt{2}$ V (B) $\sqrt{2}$ V (C) 220 V (D) zero

Section (B) : Power consumed in an AC circuit

- B-1.** The average power delivered to a series AC circuit is given by (symbols have their usual meaning) :
- (A) $E_{rms} I_{rms}$ (B) $E_{rms} I_{rms} \cos \phi$ (C) $E_{rms} I_{rms} \sin \phi$ (D) zero
- B-2.** Energy dissipates in LCR circuit in :
- (A) L only (B) C only (C) R only (D) all of these
- B-3.** The potential difference V across and the current I flowing through an instrument in an AC circuit are given by :
- $V = 5 \cos \omega t$ volt
 $I = 2 \sin \omega t$ Amp.
- The power dissipated in the instrument is :
- (A) zero (B) 5 watt (C) 10 watt (D) 2.5 watt
- B-4.** A direct current of 2 A and an alternating current having a maximum value of 2 A flow through two identical resistances. The ratio of heat produced in the two resistances in the same time interval will be:
- (A) 1 : 1 (B) 1 : 2 (C) 2 : 1 (D) 4 : 1
- B-5.** A sinusoidal AC current flows through a resistor of resistance R . If the peak current is I_p , then average power dissipated is :
- (A) $I_p^2 R \cos \theta$ (B) $\frac{1}{2} I_p^2 R$ (C) $\frac{4}{\pi} I_p^2 R$ (D) $\frac{1}{\pi^2} I_p^2 R$
- B-6.** What is the rms value of an alternating current which when passed through a resistor produces heat, which is thrice that produced by a D.C. current of 2 ampere in the same resistor in the same time interval?
- (A) 6 ampere (B) 2 ampere (C) $2\sqrt{3}$ ampere (D) 0.65 ampere





- B-7.** A resistor and a capacitor are connected to an AC supply of 200 volt, 50 Hz in series. The current in the circuit is 2 ampere. If the power consumed in the circuit is 100 watt, then the resistance in the circuit is:
 (A) 100 Ω (B) 25 Ω (C) $\sqrt{125 \times 75}$ Ω (D) 400 Ω
- B-8.** The impedance of a series circuit consists of 3 ohm resistance and 4 ohm reactance. The power factor of the circuit is :
 (A) 0.4 (B) 0.6 (C) 0.8 (D) 1.0
- B-9.** A coil of inductance 5.0 mH and negligible resistance is connected to an alternating voltage $V = 10 \sin(100t)$. The peak current in the circuit will be :
 (A) 2 amp (B) 1 amp (C) 10 amp (D) 20 amp
- B-10.** An electric bulb and a capacitor are connected in series with an AC source. On increasing the frequency of the source, the brightness of the bulb :
 (A) increase (B) decreases
 (C) remains unchanged (D) sometimes increases and sometimes decreases
- B-11.** By what percentage the impedance in an AC series circuit should be increased so that the power factor changes from $(1/2)$ to $(1/4)$ (when R is constant) ?
 (A) 200% (B) 100% (C) 50% (D) 400%
- B-12.** If the frequency of the source e.m.f. in an AC circuit is n , the power varies with a frequency :
 (A) n (B) $2n$ (C) $n/2$ (D) zero

Section (C) : AC source with R, L, C connected in series

- C-1.** A 0.21-H inductor and a 88- Ω resistor are connected in series to a 220-V, 50-Hz AC source. The current in the circuit and the phase angle between the current and the source voltage are respectively.
 (Use $\pi = 22/7$)
 (A) 2 A, $\tan^{-1} 3/4$ (B) 14.4 A, $\tan^{-1} 7/8$ (C) 14.4 A, $\tan^{-1} 8/7$ (D) 3.28 A, $\tan^{-1} 2/11$
- C-2.** A 100 volt AC source of angular frequency 500 rad/s is connected to a LCR circuit with $L = 0.8$ H, $C = 5 \mu\text{F}$ and $R = 10 \Omega$, all connected in series. The potential difference across the resistance is
 (A) $\frac{100}{\sqrt{2}}$ volt (B) 100 volt (C) 50 volt (D) $50\sqrt{3}$
- C-3.** A pure resistive circuit element X when connected to an AC supply of peak voltage 200 V gives a peak current of 5 A which is in phase with the voltage. A second circuit element Y, when connected to the same AC supply also gives the same value of peak current but the current lags behind by 90° . If the series combination of X and Y is connected to the same supply, what will be the rms value of current ?
 (A) $\frac{10}{\sqrt{2}}$ amp (B) $\frac{5}{\sqrt{2}}$ amp (C) $\frac{5}{2}$ amp (D) 5 amp
- C-4.** In an AC circuit, a resistance of R ohm is connected in series with an inductance L. If phase angle between voltage and current be 45° , the value of inductive reactance will be.
 (A) $R/4$ (B) $R/2$
 (C) R (D) cannot be found with the given data
- C-5.** In an AC circuit the potential differences across an inductor and resistor joined in series are respectively 16 V and 20 V. The total potential difference across the circuit is
 (A) 20 V (B) 25.6 V (C) 31.9 V (D) 53.5 V

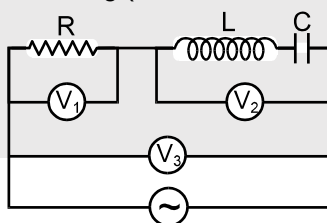




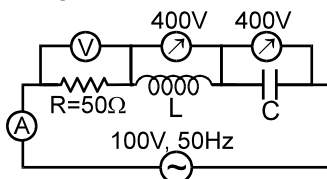
- C-6.** An AC voltage source $V = 200\sqrt{2} \sin 100t$ is connected across a circuit containing an AC ammeter (it reads rms value) and capacitor of capacity $1 \mu\text{F}$. The reading of ammeter is :
 (A) 10 mA (B) 20 mA (C) 40 mA (D) 80 mA
- C-7.** When 100 V DC is applied across a solenoid, a steady current of 1 A flows in it. When 100 V AC is applied across the same solenoid, the current drops to 0.5 A. If the frequency of the AC source is $150/\sqrt{3} \pi \text{ Hz}$, the impedance and inductance of the solenoid are :
 (A) 200Ω and $1/3 \text{ H}$ (B) 100Ω and $1/16 \text{ H}$ (C) 200Ω and 1.0 H (D) 1100Ω and $3/117 \text{ H}$
- C-8.** If in a series LCR AC circuit, the rms voltage across L, C and R are V_1 , V_2 and V_3 respectively, then the voltage of the source is always :
 (A) equal to $V_1 + V_2 + V_3$ (B) equal to $V_1 - V_2 + V_3$
 (C) more than $V_1 + V_2 + V_3$ (D) none of these is true

Section (D) : Resonance

- D-1.** The value of power factor $\cos\phi$ in series LCR circuit at resonance is :
 (A) zero (B) 1 (C) $1/2$ (D) $1/2 \text{ ohm}$
- D-2.** A series LCR circuit containing a resistance of 120 ohm has angular resonance frequency $4 \times 10^3 \text{ rad s}^{-1}$. At resonance, the voltage across resistance and inductance are 60V and 40 V respectively. The values of L and C are respectively :
 (A) 20 mH , $25/8 \mu\text{F}$ (B) 2mH , $1/35 \mu\text{F}$ (C) 20 mH , $1/40 \mu\text{F}$ (D) 2mH , $25/8 \text{ nF}$
- D-3.** In an LCR circuit, the capacitance is made one-fourth, when in resonance. Then what should be the change in inductance, so that the circuit remains in resonance ?
 (A) 4 times (B) $1/4$ times (C) 8 times (D) 2 times
- D-4.** A resistor R, an inductor L and a capacitor C are connected in series to an oscillator of frequency ν . If the resonant frequency is ν_r , then the current lags behind voltage, when :
 (A) $\nu = 0$ (B) $\nu < \nu_r$ (C) $\nu = \nu_r$ (D) $\nu > \nu_r$
- D-5.** A resistor R, an inductor L, a capacitor C and voltmeters V_1 , V_2 and V_3 are connected to an oscillator in the circuit as shown in the adjoining diagram. When the frequency of the oscillator is increased, upto resonance frequency, the voltmeter reading (at resonance frequency) is zero in the case of :



- (A) voltmeter V_1 (B) voltmeter V_2 (C) voltmeter V_3 (D) all the three voltmeters
- D-6.** In the series LCR circuit as shown in figure, the voltmeter and ammeter readings are :



- (A) $V = 100 \text{ volt}$, $I = 2 \text{ amp}$ (B) $V = 100 \text{ volt}$, $I = 5 \text{ amp}$
 (C) $V = 1000 \text{ volt}$, $I = 2 \text{ amp}$ (D) $V = 300 \text{ volt}$, $I = 1 \text{ amp}$



Section (E) : Transformer

- E-1.** A power transformer (step up) with an 1 : 8 turn ratio has 60 Hz, 120 V across the primary; the load in the secondary is $10^4 \Omega$. The current in the secondary is
 (A) 96 A (B) 0.96 A (C) 9.6 A (D) 96 mA
- E-2.** A transformer is used to light a 140 watt, 24 volt lamp from 240 V AC mains. The current in the main cable is 0.7 amp. The efficiency of the transformer is :
 (A) 48% (B) 63.8% (C) 83.3% (D) 90%
- E-3.** In a step-up transformer the voltage in the primary is 220 V and the current is 5A. The secondary voltage is found to be 22000 V. The current in the secondary (neglect losses) is
 (A) 5 A (B) 50 A (C) 500 A (D) 0.05 A
- E-4.** The core of a transformer is laminated to reduce
 (A) eddy current loss (B) hysteresis loss (C) copper loss (D) magnetic loss

Section (F) : Miscellaneous

- F-1.** A capacitor is a perfect insulator for :
 (A) constant direct current (B) alternating current
 (C) direct as well as alternating current (D) variable direct current
- F-2.** A choke coil should have :
 (A) high inductance and high resistance (B) low inductance and low resistance
 (C) high inductance and low resistance (D) low inductance and high resistance
- F-3.** A choke coil is preferred to a rheostat in AC circuit as :
 (A) it consumes almost zero power (B) it increases current
 (C) it increases power (D) it increases voltage
- F-4.** With increase in frequency of an AC supply, the inductive reactance :
 (A) decreases (B) increases directly proportional to frequency
 (C) increases as square of frequency (D) decreases inversely with frequency
- F-5.** With increase in frequency of an AC supply, the capacitive reactance :
 (A) varies inversely with frequency (B) varies directly with frequency
 (C) varies directly as square of frequency (D) remains constant
- F-6.** An AC ammeter is used to measure current in a circuit. When a given direct constant current passes through the circuit, the AC ammeter reads 3 ampere. When an alternating current passes through the circuit, the AC ammeter reads 4 ampere. Then the reading of this ammeter if DC and AC flow through the circuit simultaneously, is :
 (A) 3 A (B) 4 A (C) 7 A (D) 5 A
- F-7.** In an a.c. circuit consisting of resistance R and inductance L, the voltage across R is 60 volt and that across L is 80 volt. The total voltage across the combination is
 (A) 140 V (B) 20 V (C) 100 V (D) 70 V

PART - III : MATCH THE COLUMN

1. Match the Physical quantities given in column-I with the parameters they depend on as given in column-II.

Column I

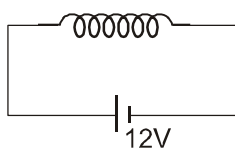
- (A) Inductance of a coil
 (B) Capacitance
 (C) Impedance of a coil
 (D) Reactance of a capacitor

Column II

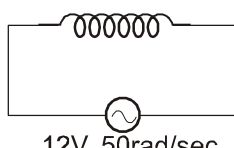
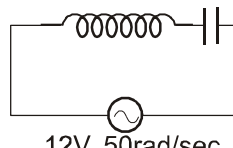
- (p) Depends on resistivity
 (q) Depends on shape
 (r) Depends on medium inserted
 (s) Depends on external AC voltage source



2. A steady current 4 A flows in an inductor coil when connected to a 12 V dc source as shown in figure 1. If the same coil is connected to an ac source of 12 V, 50 rad/s, a current of 2.4 A flows in the circuit as shown in figure 2. Now after these observations, a capacitor of capacitance $\frac{1}{50}$ F is connected in series with the coil as shown in figure 3 with the same AC source :



(Figure 1)


 12V, 50rad/sec.
(Figure 2)

 12V, 50rad/sec.
(Figure 3)

Column-I

- (A) The inductance of the coil (nearly)
(B) The resistance of the coil
(C) Average power (nearly)
(D) Total reactance

Column-II (in S.I units)

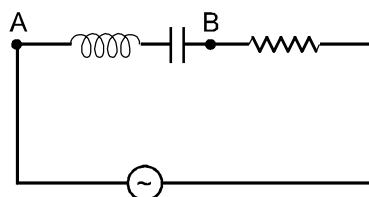
- (p) 24
(q) 3
(r) 0.08

Exercise-2

Marked Questions can be used as Revision Questions.

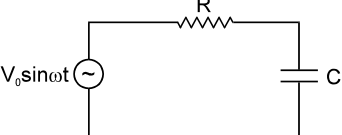
PART - I : ONLY ONE OPTION CORRECT TYPE

- A coil has an inductance of $\frac{2.2}{\pi}$ H and is joined in series with a resistance of 220Ω . When an alternating e.m.f. of 220 V at 50 c.p.s. is applied to it, then the wattless component of the rms current in the circuit is
(A) 5 ampere (B) 0.5 ampere (C) 0.7 ampere (D) 7 ampere
- The current in a circuit containing a capacitance C and a resistance R in series leads over the applied voltage of frequency $\frac{\omega}{2\pi}$ by.
(A) $\tan^{-1}\left(\frac{1}{\omega CR}\right)$ (B) $\tan^{-1}(\omega CR)$ (C) $\tan^{-1}\left(\omega \frac{1}{R}\right)$ (D) $\cos^{-1}(\omega CR)$ [REE - 1991]
- An inductor $\left(L = \frac{1}{100\pi} \text{ H}\right)$, a capacitor $\left(C = \frac{1}{500\pi} \text{ F}\right)$ and a resistance (3Ω) is connected in series with an AC voltage source as shown in the figure. The voltage of the AC source is given as $V = 10 \cos(100\pi t)$ volt. What will be the potential difference between A and B ?



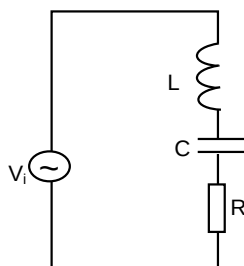
- (A) $8 \cos(100\pi t - 127^\circ)$ volt (B) $8 \cos(100\pi t - 53^\circ)$ volt
(C) $8 \cos(100\pi t - 37^\circ)$ volt (D) $8 \cos(100\pi t + 37^\circ)$ volt



4. An ac voltage source $V = V_0 \sin \omega t$ is connected across resistance R and capacitance C as shown in figure. It is given that $R = \frac{1}{\omega C}$. The peak current is I_0 . If the angular frequency of the voltage source is changed to $\frac{\omega}{\sqrt{3}}$ keeping R and C fixed, then the new peak current in the circuit is :
- 
- (A) $\frac{I_0}{2}$ (B) $\frac{I_0}{\sqrt{2}}$ (C) $\frac{I_0}{\sqrt{3}}$ (D) $\frac{I_0}{3}$
5. For a LCR series circuit with an A.C. source of angular frequency ω .
- (A) circuit will be capacitive if $\omega > \frac{1}{\sqrt{LC}}$ (B) circuit will be inductive if $\omega = \frac{1}{\sqrt{LC}}$
- (C) power factor of circuit will be unity if capacitive reactance equals inductive reactance
- (D) current will be leading voltage if $\omega > \frac{1}{\sqrt{LC}}$
6. An LCR series circuit with 100Ω resistance is connected to an AC source of 200 V and angular frequency 300 radians per second. When only the capacitance is removed, the current lags the voltage by 60° . When only the inductance is removed, the current leads the voltage by 60° . Then the current and power dissipated in LCR circuit are respectively
- (A) 1 A , 200 watt . (B) 1 A , 400 watt . (C) 2 A , 200 watt . (D) 2 A , 400 watt .
7. In an L-R series circuit ($L = \frac{175}{11} \text{ mH}$ and $R = 12 \Omega$), a variable emf source ($V = V_0 \sin \omega t$) of $V_{\text{rms}} = 130\sqrt{2} \text{ V}$ and frequency 50 Hz is applied. The current amplitude in the circuit and phase of current with respect to voltage are respectively (Use $\pi = 22/7$)
- (A) 14.14 A , 30° (B) $10\sqrt{2} \text{ A}$, $\tan^{-1} \frac{5}{12}$ (C) 10 A , $\tan^{-1} \frac{5}{12}$ (D) 20 A , $\tan^{-1} \frac{5}{12}$
8. In LCR circuit at resonance current in the circuit is $10\sqrt{2} \text{ A}$. If now frequency of the source is changed such that now current lags by 45° than applied voltage in the circuit. Which of the following is correct :
- (A) Frequency must be increased and current after the change is 10 A
- (B) Frequency must be decreased and current after the change is 10 A
- (C) Frequency must be decreased and current is same as that of initial value
- (D) The given information is insufficient to conclude anything
9. The overall efficiency of a transformer is 90% . The transformer is rated for an output of 9000 watt . The primary voltage is 1000 volt . The ratio of turns in the primary to the secondary coil is $5 : 1$. The iron losses at full load are 700 watt . The primary coil has a resistance of 1 ohm .
- (i) The voltage in secondary coil is :
- (A) 1000 volt (B) 5000 volt (C) 200 volt (D) zero volt
- (ii) In the above, the current in the primary coil is :
- (A) 9 amp (B) 10 amp (C) 1 amp (D) 4.5 amp
- (iii) In the above, the copper loss in the primary coil is :
- (A) 100 watt (B) 700 watt (C) 200 watt (D) 1000 watt
- (iv) In the above, the copper loss in the secondary coil is :
- (A) 100 watt (B) 700 watt (C) 200 watt (D) 1000 watt
- (v) In the above, the current in the secondary coil is :
- (A) 45 amp (B) 46 amp (C) 10 amp (D) 50 amp
- (vi) In the above, the resistance of the secondary coil is approximately :
- (A) 0.01Ω (B) 0.1Ω (C) 0.2Ω (D) 0.4Ω



10. In the following circuit the current is in phase with the applied voltage. Therefore, the current in the circuit and the frequency of the source voltage respectively, are [Olympiad (Stage-1) 2017]

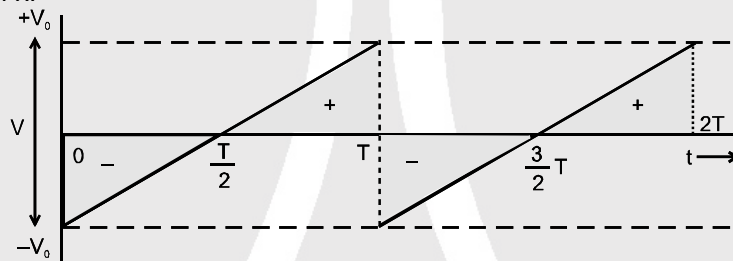


- (A) $\frac{V_i}{R}$ and $\frac{1}{2\pi\sqrt{LC}}$ (B) zero and $\frac{1}{\sqrt{LC}}$ (C) $\sqrt{\frac{C}{L}}V_i$ and $\frac{2}{\pi\sqrt{LC}}$ (D) $4\sqrt{\frac{C}{LR^2}}$ and $\frac{2}{\sqrt{LC}}$

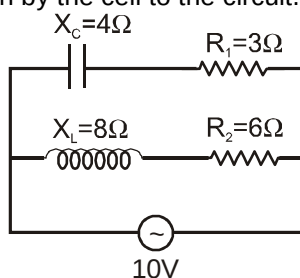
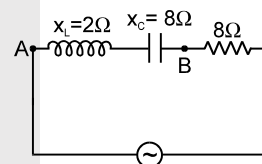
PART - II : SINGLE AND DOUBLE VALUE INTEGER TYPE

1. The rms value for the saw-tooth voltage of peak value V_0 from $t = 0$ to $t = 2T$ as shown in figure is $\frac{V_0}{\sqrt{x}}$.

Find the value of x .



2. An inductor ($x_L = 2\Omega$) a capacitor ($x_C = 8\Omega$) and a resistance (8Ω) is connected in series with an ac source. The voltage output of A.C source is given by $v = 10 \cos 100\pi t$. The instantaneous p.d. between A and B is equal to $x \times 10^{-1}$ volt, when it is half of the voltage output from source at that instant. Find out value of x .
3. A 2000 Hz, 20 volt source is connected to a resistance of 20 ohm, an inductance of $0.125/\pi$ H and a capacitance of $500/\pi$ nF all in series. Calculate the time (in seconds) in which the resistance (thermal capacity = 100 joule/ $^\circ\text{C}$) will get heated by 10°C . (Assume no loss of heat)
4. A series LCR circuit containing a resistance of 120 ohm has angular resonance frequency $4 \times 10^5 \text{ rad s}^{-1}$. At resonance, the voltage across resistance and inductance are 60V and 40 V respectively. At frequency the current in the circuit lags the voltage by 45° is equal to $x \times 10^5 \text{ rad/sec}$. Find value of x [REE - 1995]
5. An LCR circuit has $L = 10 \text{ mH}$, $R = 150 \Omega$ and $C = 1 \mu\text{F}$ connected in series to a source of $150\sqrt{2} \cos \omega t$ volt. At a frequency that is 50% of the resonant frequency, calculate the average power (in watt) dissipated per cycle.
6. In the figure shown an ideal alternative current (A.C.) source of 10 Volt is connected. Find half of the total average power (in watts) given by the cell to the circuit.




PART - III : ONE OR MORE THAN ONE OPTIONS CORRECT TYPE

- Average power consumed in an A.C. series circuit is given by (symbols have their usual meaning) :

(A) $E_{rms} I_{rms} \cos \phi$ (B) $(I_{rms})^2 R$ (C) $\frac{E_{max}^2 R}{2(|z|)^2}$ (D) $\frac{I_{max}^2 |z| \cos \phi}{2}$
- An AC source supplies a current of 10 A (rms) to a circuit, rms voltage of source is 100 V. The average power delivered by the source :

(A) must be 1000 W (B) may be less than 1000 W
(C) may be greater than 1000 W (D) may be 1000 W
- Which of the following quantities have zero average value over a cycle. If an inductor coil having some resistance is connected to a sinusoidal AC source.

(A) induced emf in the inductor (B) current
(C) joule heat (D) magnetic energy stored in the inductor
- In a series LCR circuit with an AC source ($E_{rms} = 50$ V and $\nu = 50/\pi$ Hz), $R = 300 \Omega$, $C = 0.02$ mF, $L = 1.0$ H, which of the following is correct

(A) the rms current in the circuit is 0.1 A
(B) the rms potential difference across the capacitor is 50 V
(C) the rms potential difference across the capacitor is 14.1 V
(D) the rms current in the circuit is 0.14 A
- In an AC series circuit when the instantaneous source voltage is maximum, the instantaneous current is zero. Connected to the source may be a

(A) pure capacitor
(B) pure inductor
(C) combination of pure an inductor and pure capacitor
(D) pure resistor
- A coil of inductance 5.0 mH and negligible resistance is connected to an oscillator giving an output voltage $E = (10V) \sin \omega t$. Which of the following is correct

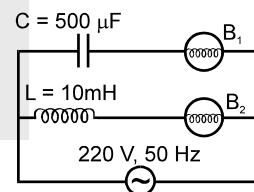
(A) for $\omega = 100$ s⁻¹ peak current is 20 A (B) for $\omega = 500$ s⁻¹ peak current is 4 A
(C) for $\omega = 1000$ s⁻¹ peak current is 2 A (D) for $\omega = 1000$ s⁻¹ peak current is 4 A
- A pure inductance of 1 henry is connected across a 110 V, 70Hz source. Then correct option are (Use $\pi = 22/7$):

(A) reactance of the circuit is 440 Ω (B) current of the circuit is 0.25 A
(C) reactance of the circuit is 880 Ω (D) current of the circuit is 0.5 A
- In the circuit shown in figure, if both the bulbs B_1 and B_2 are identical :

(A) their brightness will be the same
(B) B_2 will be brighter than B_1
(C) as frequency of supply voltage is increased the brightness of bulb B_1 will increase and that of B_2 will decrease.
(D) only B_2 will glow because the capacitor has infinite impedance
- A circuit is set up by connecting $L = 100$ mH, $C = 5$ μ F and $R = 100 \Omega$ in series. An alternating emf of $(150\sqrt{2})$ volt, $\frac{500}{\pi}$ Hz is applied across this series combination. Which of the following is correct

(A) the impedance of the circuit is 141.4 Ω
(B) the average power dissipated across resistance 225 W
(C) the average power dissipated across inductor is zero.
(D) the average power dissipated across capacitor is zero.
- In a series RC circuit with an AC source (peak voltage $E_0 = 50$ V and $f = 50/\pi$ Hz), $R = 300 \Omega$, $C = 25$ μ F. Then

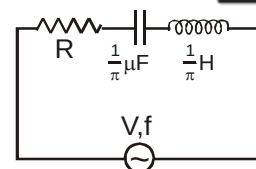
(A) the peak current is 0.1 A (B) the peak current is 0.7 A
(C) the average power dissipated is 1.5 W (D) the average power dissipated is 3 W



Alternating Current



11. In the AC circuit shown below, the supply voltage has constant rms value V but variable frequency f . At resonance, the circuit :



- (A) has a current I given by $I = V/R$
 (B) has a resonance frequency 500 Hz
 (C) has a voltage across the capacitor which is 180° out of phase with that across the inductor

(D) has a current given by $I = \frac{V}{\sqrt{R^2 + \left(\frac{1}{\pi} + \frac{1}{\pi}\right)^2}}$

12. A town situated 20 km away from a power house at 440 V, requires 600 kW of electric power at 220 V. The resistance of transmission line carrying power is 0.4Ω per km. The town gets power from the line through a 3000 V–220 V step-down transformer at a substation in the town. Which of the following is/are correct

- (A) The loss in the form of heat is 640 kW (B) The loss in the form of heat is 1240 kW
 (C) Plant should supply 1240 kW (D) Plant should supply 640 kW

13. 11 kW of electric power can be transmitted to a distant station at (i) 220 V or (ii) 22000 V. Which of the following is correct

- (A) first mode of transmission consumes less power (B) second mode of transmission consumes less power
 (C) first mode of transmission draws less current (D) second mode of transmission draws less current

14. Power factor may be equal to 1 for :

- (A) pure inductor (B) pure capacitor (C) pure resistor (D) An LCR circuit

15. In a series R-C circuit the supply voltage (V_s) is kept constant at 2V and the frequency f of the sinusoidal voltage is varied from 500 Hz to 2000 Hz. The voltage across the resistance $R = 1000 \Omega$ is measured each time as V_R . For the determination of the C a student wants to draw a linear graph and try to get C from the slope. Then she may draw a graph of

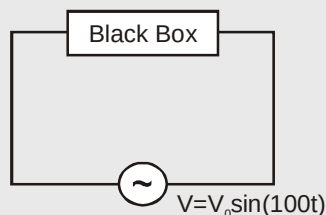
[Olympiad (Stage-1) 2017]

- (A) f^2 against V_R^2 (B) $\frac{1}{f^2}$ against $\frac{V_s^2}{V_R^2}$ (C) $\frac{1}{f^2}$ against $\frac{1}{V_R^2}$ (D) f against $\frac{V_R}{\sqrt{V_s^2 - V_R^2}}$

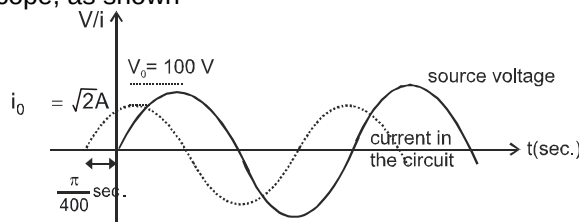
PART - IV : COMPREHENSION

Comprehension-1

A voltage source $V = V_0 \sin(100t)$ is connected to a black box in which there can be either one element out of L, C, R or any two of them connected in series.



At steady state the variation of current in the circuit and the source voltage are plotted together with time, using an oscilloscope, as shown



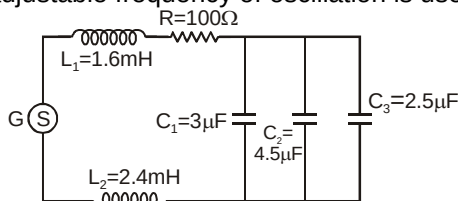
1. The element(s) present in black box is/are :
 (A) only C (B) L and C (C) L and R (D) R and C
2. Values of the parameters of the elements, present in the black box are -
 (A) $R = 50 \Omega$, $C = 200 \mu F$ (B) $R = 50 \Omega$, $L = 2 mH$ (C) $R = 400 \Omega$, $C = 50 \mu F$ (D) None of these



3. If AC source is removed, the circuit is shorted for some time so that capacitor is fully discharged and then a battery of constant EMF is connected across the black box, at $t = 0$. The current in the circuit will
- increase exponentially with time constant = 0.02 sec.
 - decrease exponentially with time constant = 0.01 sec.
 - oscillate with angular frequency 20 rad/sec
 - first increase and then decrease

Comprehension-2

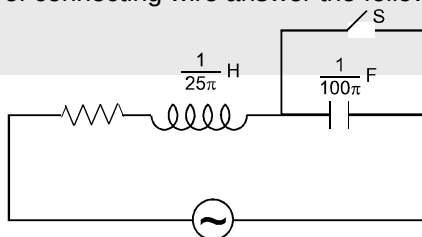
An ac generator G with an adjustable frequency of oscillation is used in the circuit, as shown.



4. Current drawn from the ac source will be maximum if its angular frequency is -
 (A) 10^5 rad/s (B) 10^4 rad/s (C) 5000 rad/s (D) 500 rad/s
5. To increase resonant frequency of the circuit, some of the changes in the circuit are carried out. Which change(s) would certainly result in the increase in resonant frequency ?
 (A) R is increased. (B) L_1 is increased and C_1 is decreased.
 (C) L_2 is decreased and C_2 is increased. (D) C_3 is removed from the circuit.
6. If the ac source G is of 100 V rating at resonant frequency of the circuit, then average power supplied by the source is -
 (A) 50 W (B) 100 W (C) 500 W (D) 1000 W
7. Average energy stored by the inductor L_2 (Source is at resonance frequency) is equal to
 (A) zero (B) 1.2 mJ (C) 2.4 mJ (D) 4 mJ
8. Thermal energy produced by the resistance R in time duration 1 μ s, using the source at resonant condition, is
 (A) 0 J (B) 1 μ J
 (C) 100 μ J (D) not possible to calculate from the given information

Comprehension-3

In the LCR circuit shown in figure unknown resistance and alternating voltage source are connected. When switch 'S' is closed then there is a phase difference of $\frac{\pi}{4}$ between current and applied voltage and voltage across resistor is $\frac{100}{\sqrt{2}}$ V. When switch is open current and applied voltage are in same phase. Neglecting resistance of connecting wire answer the following questions :



9. Peak voltage of applied voltage sources is :
 (A) $200\sqrt{2}$ V (B) 100 V (C) $100\sqrt{2}$ V (D) $\frac{100}{\sqrt{2}}$ V
10. Resonance frequency of circuit is :
 (A) 50 Hz (B) 25 Hz
 (C) 75 Hz (D) Data insufficient for calculation
11. Average power consumption in the circuit when 'S' is open :
 (A) 2500 W (B) 3000 W (C) 5000 W (D) 1250 W



Exercise-3

Marked Questions can be used as Revision Questions.

* Marked Questions may have more than one correct option.

PART - I : JEE (ADVANCED) / IIT-JEE PROBLEMS (PREVIOUS YEARS)

1. An AC voltage source of variable angular frequency ω and fixed amplitude V connected in series with a capacitance C and an electric bulb of resistance R (inductance zero). When ω is increased :

[JEE 2010; 3/163, -1]

- (A) the bulb glows dimmer (B) the bulb glows brighter
(C) total impedance of the circuit is unchanged (D) total impedance of the circuit increases

2. You are given many resistances, capacitors and inductors. These are connected to a variable DC voltage source (the first two circuits) or an AC voltage source of 50 Hz frequency (the next three circuits) in different ways as shown in **Column II**. When a current I (steady state for DC or rms for AC) flows through the circuit, the corresponding voltage V_1 and V_2 . (indicated in circuits) are related as shown in **Column I**. Match the two column.

[JEE 2010; 8/163]

Column I

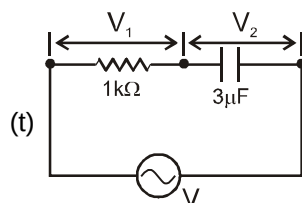
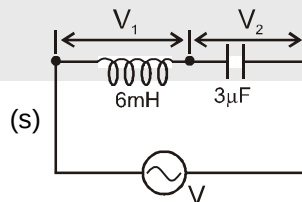
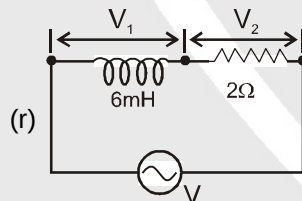
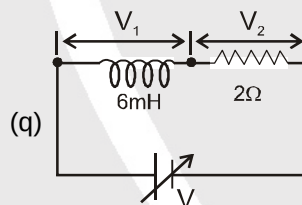
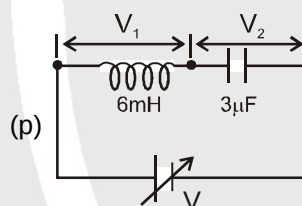
(A) $I \neq 0, V_1$ is proportional to I

(B) $I \neq 0, V_2 > V_1$

(C) $V_1 = 0, V_2 = V$

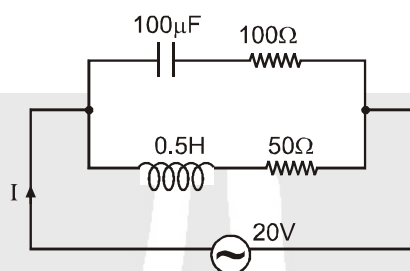
(D) $I \neq 0, V_2$ is proportional to I

Column II





- 3.* A series R-C circuit is connected to AC voltage source. Consider two cases; (A) when C is without a dielectric medium and (B) when C is filled with dielectric of constant 4. The current I_R through the resistor and voltage V_C across the capacitor are compared in the two cases. Which of the following is/are true? [JEE 2011; 4/160]
- (A) $I_R^A > I_R^B$ (B) $I_R^A < I_R^B$ (C) $V_C^A > V_C^B$ (D) $V_C^A < V_C^B$
4. A series R-C combination is connected to an AC voltage of angular frequency $\omega = 500$ radian/s. If the impedance of the R-C circuit is $R\sqrt{1.25}$, the time constant (in millisecond) of the circuit is [JEE 2011; 4/160]
- 5.* In the given circuit, the AC source has $\omega = 100$ rad/s. considering the inductor and capacitor to be ideal, the correct choice (s) is(are) [IIT-JEE-2012, Paper-2; 4/66]



- (A) The current through the circuit, I is approximately 0.3A
 (B) The current through the circuit, I is $0.3\sqrt{2}$ A.
 (C) The voltage across 100Ω resistor = $10\sqrt{2}$ V
 (D) The voltage across 50Ω resistor = 10V

Paragraph for Questions 6 and 7

A thermal power plant produces electric power of 600 kW at 4000 V, which is to be transported to a place 20 km away from the power plant for consumers' usage. It can be transported either directly with a cable of large current carrying capacity or by using a combination of step-up and step-down transformers at the two ends. The drawback of the direct transmission is the large energy dissipation. In the method using transformers, the dissipation is much smaller. In this method, a step-up transformer is used at the plant side so that the current is reduced to a smaller value. At the consumers' end, a step-down transformer is used to supply power to the consumers at the specified lower voltage. It is reasonable to assume that the power cable is purely resistive and the transformers are ideal with a power factor unity. All the currents and voltages mentioned are rms values. [JEE(Advanced) 2013; 3/60]

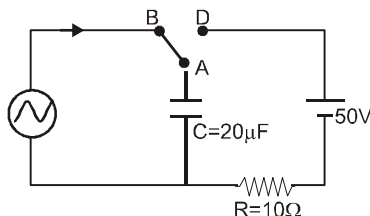
6. If the direct transmission method with a cable of resistance $0.4 \Omega \text{ km}^{-1}$ is used, the power dissipation (in %) during transmission is :
 (A) 20 (B) 30 (C) 40 (D) 60
7. In the method using the transformers, assume that the ratio of the number of turns in the primary to that in the secondary in the step-up transformer is 1 : 10. If the power to the consumers has to be supplied at 200V, the ratio of the number of turns in the primary to that in the secondary in the step-down transformer is :
 (A) 200 : 1 (B) 150 : 1 (C) 100 : 1 (D) 50 : 1





- 8.* At time $t = 0$, terminal A in the circuit shown in the figure is connected to B by a key and alternating current $I(t) = I_0 \cos(\omega t)$, with $I_0 = 1\text{A}$ and $\omega = 500\text{ rad s}^{-1}$ starts flowing in it with the initial direction shown in the figure. At $t = \frac{7\pi}{6\omega}$, the key is switched from B to D. Now onwards only A and D are connected. A total charge Q flows from the battery to charge the capacitor fully. If $C = 20\mu\text{F}$, $R = 10\Omega$ and the battery is ideal with emf of 50V , identify the correct statement (s)

[JEE (Advanced) 2014,P-1, 3/60]



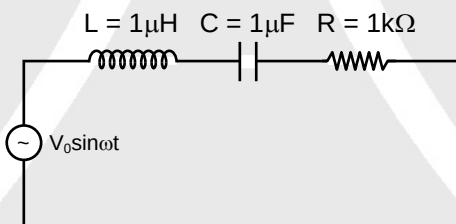
(A) Magnitude of the maximum charge on the capacitor before $t = \frac{7\pi}{6\omega}$ is $1 \times 10^{-3}\text{ C}$.

(B) The current in the left part of the circuit just before $t = \frac{7\pi}{6\omega}$ is clockwise

(C) Immediately after A is connected to D, the current in R is 10A .

(D) $Q = 2 \times 10^{-3}\text{ C}$.

- 9*. In the circuit shown, $L = 1\mu\text{H}$, $C = 1\mu\text{F}$ and $R = 1\text{k}\Omega$. They are connected in series with an a.c. source $V = V_0 \sin \omega t$ as shown. Which of the following options is/are correct ? [JEE (Advanced) 2017, P-1, 4/61, -2]



(A) The current will be in phase with the voltage if $\omega = 10^4\text{ rad.s}^{-1}$

(B) At $\omega \gg 10^6\text{ rad.s}^{-1}$, the circuit behaves like a capacitor

(C) The frequency at which the current will be in phase with the voltage is independent of R

(D) At $\omega \sim 0$ the current flowing through the circuit becomes nearly zero

- 10*. The instantaneous voltages at three terminals marked X, Y and Z are given by

$$V_X = V_0 \sin \omega t, V_Y = V_0 \sin \left(\omega t + \frac{2\pi}{3} \right) \text{ and } V_Z = V_0 \sin \left(\omega t + \frac{4\pi}{3} \right)$$

An ideal voltmeter is configured to read rms value of the potential difference between its terminals. It is connected between points X and Y and then between Y and Z. The reading (s) of the voltmeter will be

[JEE(Advanced) 2017 ; P-2, 4/61, -2]

(A) $V_{XY}^{\text{rms}} = V_0 \sqrt{\frac{3}{2}}$

(B) $V_{YZ}^{\text{rms}} = V_0 \sqrt{\frac{1}{2}}$

(C) independent of the choice of the two terminals (D) $V_{XY}^{\text{rms}} = V_0$



PART - II : JEE (MAIN) / AIEEE PROBLEMS (PREVIOUS YEARS)

1. A circuit has a resistance of 12 ohm and an impedance of 15 ohm. The power factor of the circuit will be :
 (1) 0.8 (2) 0.4 (3) 1.25 (4) 0.125
[AIEEE 2005; 4/300]
2. The phase difference between the alternating current and emf is $\pi/2$. Which of the following cannot be the constituent of the circuit?
 (1) C alone (2) R, L (3) L, C (4) L alone
[AIEEE 2005; 4/300]
3. In a series LCR circuit $R = 200 \Omega$ and the voltage and the frequency of the main supply is 220 V and 50 Hz respectively. On taking out the capacitance from the circuit the current lags behind the voltage by 30° . On taking out the inductor from the circuit the current leads the voltage by 30° . The power dissipated in the LCR circuit is
 (1) 305 W (2) 210 W (3) 0 W (4) 242 W
[AIEEE 2010; 4/144, -1]
4. An arc lamp requires a direct current of 10 A at 80 V to function. If it is connected to a 220V(rms), 50 Hz AC supply, the series inductor needed for it to work is close to :
 (1) 0.08 H (2) 0.044 H (3) 0.065 H (4) 80 H
[JEE (Main) 2016, 4/120, -1]
5. For an RLC circuit driven with voltage of amplitude v_m and frequency $\omega_0 = \frac{1}{\sqrt{LC}}$ the current exhibits resonance. The quality factor, Q is given by :
 (1) $\frac{R}{\omega_0 C}$ (2) $\frac{CR}{\omega_0}$ (3) $\frac{\omega_0 L}{R}$ (4) $\frac{\omega_0 R}{L}$
[JEE (Main) 2018; 4/120, -1]
6. In an a.c circuit, the instantaneous e.m.f and current are given by
 $e = 100 \sin 30t$
 $i = 20 \sin \left(30t - \frac{\pi}{4} \right)$
 In one cycle of a.c the average power consumed by the circuit and the wattless current are, respectively:
 (1) $\frac{50}{\sqrt{2}}, 0$ (2) 50, 0 (3) 50, 10 (4) $\frac{1000}{\sqrt{2}}, 10$
[JEE (Main) 2018; 4/120, -1]

Answers

EXERCISE-1

PART - I

Section (A) :

A-1. $220\sqrt{2}$ V, 2.5 ms A-2. $\frac{I_0}{e} \sqrt{(e^2 - 1)/2}$

A-3. $I_{\text{eff}} = \left[a^2 + \frac{1}{2}b^2 \right]^{1/2}$ A-4. 0

Section (B) :

B-1. $12\sqrt{2}$ volts B-2. 0.72 W

B-3. 10 W, 5 W

Section (C) :

C-1. 3.0 kV

C-2. $\frac{2.2\sqrt{3}}{\pi} = 1.2 \text{ H} = \frac{7\sqrt{3}}{10} \text{ H}$

C-3. (a) 0.1 A (b) 50 V, 30 V, 10 V
 (Note that the sum of the rms potential differences across the three elements is greater than the rms voltage of the source.)

C-4. 25 mJ, 5mJ

C-5. (i) $\frac{125}{33} \mu\text{F}$ or 2.4 H (ii) 720 Ω

(iii) It will be more economical to use inductance or capacitance in series with the lamp to run it as it consumes no power while there would be dissipation of power when resistance is inserted in series with the lamp.



C-6. $Z = 50\sqrt{2}$ ohm, $V_C = 500\sqrt{2}$ volt and

$$V_L = 600\sqrt{2} \text{ volt}, \frac{1}{\sqrt{2}}$$

C-7. $\sqrt{\frac{21}{22}}$ H, 100 Ω

C-8. 125 Ω , 288J

Section (D) :

D-1. $4\mu\text{F}$, $R = \frac{141.4}{5}\Omega$

D-2. (a) $\frac{250}{3\pi}$ Hz (b) 2 mA **D-3.** 1.5 A

D-4. 1×10^{-8} henry

Section (E) :

E-1 zero

E-2. (a) 800 V

(b) (i) 0.25 A (ii) 0.2 A.

PART - II

Section (A) :

A-1. (B) **A-2.** (B) **A-3.** (C)

A-4. (D) **A-5.** (D) **A-6.** (D)

Section (B) :

B-1. (B) **B-2.** (C) **B-3.** (A)

B-4. (C) **B-5.** (B) **B-6.** (C)

B-7. (B) **B-8.** (B) **B-9.** (D)

B-10. (A) **B-11.** (B) **B-12.** (B)

Section (C) :

C-1. (A) **C-2.** (B) **C-3.** (C)

C-4. (C) **C-5.** (B) **C-6.** (B)

C-7. (C) **C-8.** (D)

Section (D) :

D-1. (B) **D-2.** (A) **D-3.** (A)

D-4. (D) **D-5.** (B) **D-6.** (A)

Section (E) :

E-1. (D) **E-2.** (C) **E-3.** (D)

E-4. (A)

Section (F) :

F-1. (A) **F-2.** (C) **F-3.** (A)

F-4. (B) **F-5.** (A) **F-6.** (D)

F-7. (C)

PART - III

1. (A) $\rightarrow$ q,r; (B) $\rightarrow$ q,r; (C) $\rightarrow$ p,q,r,s; (D) $\rightarrow$ q,r, s

2. (A) $\rightarrow$ r ; (B) $\rightarrow$ q ; (C) $\rightarrow$ p ; (D) $\rightarrow$ q

EXERCISE-2

PART - I

1. (B) 2. (A) 3. (C)

4. (B) 5. (C) 6. (D)

7. (D) 8. (A)

9. (i) (C) (ii) (B) (iii) (A) (iv) (C) (v) (B)
(vi) (B)

10. (A)

PART - II

1. 3 2. 48 3. 50

4. 8 5. 75 6. 9

PART - III

1. (ABCD) 2. (BD) 3. (AB)

4. (AB) 5. (ABC) 6. (ABC)

7. (AB) 8. (BC) 9. (ABCD)

10. (AC) 11. (ABC) 12. (AC)

13. (BD) 14. (CD) 15. (BCD)

PART - IV

1. (D) 2. (A) 3. (B)

4. (C) 5. (D) 6. (B)

7. (B) 8. (D) 9. (C)

10. (B) 11. (C)

EXERCISE-3

PART - I

1. (B)

2. (A) $\rightarrow$ r,s,t; (B) $\rightarrow$ q,r,s,t; (C) $\rightarrow$ p,q; (D) $\rightarrow$ q,r,s,t

3. (BC) 4. 4 5. (AC)

6. (B) 7. (A) 8. (CD)

9. (CD) 10. (AC)

PART - II

1. (1) 2. (2) 3. (4)

4. (3) 5. (3) 6. (4)